Michael McPhee

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Technical Skills

Programming: Python (NumPy, pandas, scikit-learn, OpenCV), MATLAB, Simulink, C/C++, VHDL, git, Linux
Machine Learning: Supervised and Unsupervised Learning, Neural Networks, Ensemble Methods, Model Evaluation
Controls: PID, Kalman Filtering, System Identification
Embedded: STM32, Arduino, FPGA, RTOS, I^2C, SPI, UART
Design: SolidWorks, AutoCAD, 3D Printing, Machining, Soldering

Experience

Mission Operations Engineering Student Jan 2026 – Aug 2026
MDA Space *Brampton, ON*

- Tested and troubleshot Canadarm2 flight camera hardware returned from orbit using an oscilloscope, breakout box, and variable DC power supply
- Wrote a Python script to automate oscilloscope data collection for full test runs, then analyzed results in MATLAB and flagged anomalies to support the investigation
- Led a study on electrolytic capacitor aging in the Canadarm2 Robotic Workstation video monitors, assessing the risk to on-orbit spares and presenting a swap recommendation to the Canadian Space Agency
- Initiated procurement of all parts and subcontracted assembly of cable harnesses and circuit card assemblies for two camera development projects

Robotics Engineering May 2025 – Aug 2025
Low Power Futures *Markham, ON*

- Designed and built a two-wheeled differential-drive robot, modeling it in SolidWorks and 3D printing the chassis, drivetrain, and electronics mounts
- Wired and soldered the battery pack, custom I^2C breakout board, sensors, and motor drivers
- Wrote Arduino firmware for motor control, sensor reads, and PID loops to validate motion
- Modeled the robot's dynamics in MATLAB and implemented a Kalman filter to estimate position and orientation

Engineering Research Assistant Sep 2024 – Dec 2024
University of Waterloo Ideas Clinic *Waterloo, ON*

- Built an educational mass-spring-damper system with an ultrasonic sensor and motor encoder, plotting real-time data in Python over UART from an Arduino
- Redesigned a conveyor gearbox in SolidWorks, extending runtime from under 10 minutes to continuous operation

Engineering Co-op Jan 2024 – Apr 2024
Research Grade Sports *Harrogate, UK*

- Developed an OpenCV Python program to measure golf ball speed and spin from video, using a Raspberry Pi camera on an NVIDIA Jetson
- Modeled golf ball rolling in MATLAB, tuning parameters against experimental data to reach 0.31 m RMS error

Projects

Ball-Balancing Stewart Platform | *PID Control, AprilTags, Arduino, SolidWorks* Dec 2025

- Developed a vision-based controller for a 3-RRS Stewart platform that balances a ping pong ball, projecting ball error onto each motor axis to drive three independent PID loops without inverse kinematics
- Automated platform calibration with AprilTag pose estimation, cutting maximum calibration error from 6.89 mm to 1.60 mm (77%) across five trials

STM32 Multithreaded Real-Time Operating System | *C, RTOS, STM32* Aug 2024

- Implemented a round-robin thread scheduler in C on an STM32, with stack management and context switching for multiple user threads
- Built and tested kernel functions for thread creation, interrupt handling, and priority-based scheduling

Education

University of Waterloo Sep 2022 – Apr 2027
Mechatronics Engineering, Artificial Intelligence Option *Waterloo, ON*

- Achieved a 91% cumulative GPA, earning Dean's Honours List in Terms 2A and 2B and Term Distinction in 3A